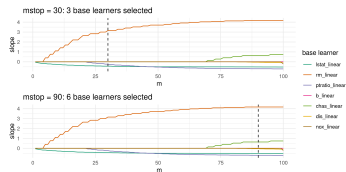
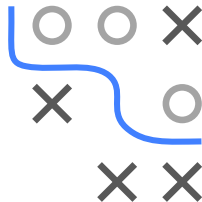


Introduction to Machine Learning

Boosting

Gradient Boosting: CWB Basics 2



Learning goals

- Handling of categorical features
- Intercept handling
- Practical example

HANDLING OF CATEGORICAL FEATURES

Feature x_j with G categories. Two options for encoding:

- One base learner to simultaneously estimate all categories:

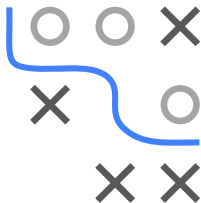
$$b_j(x_j|\theta_j) = \sum_{g=1}^G \theta_{j,g} \mathbb{1}_{\{g=x_j\}} = (\mathbb{1}_{\{x_j=1\}}, \dots, \mathbb{1}_{\{x_j=G\}}) \theta_j$$

Hence, b_j incorporates a one-hot encoded feature with group means $\theta \in \mathbb{R}^G$ as estimators.

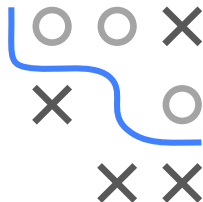
- One binary base learner per category:

$$b_{j,g}(x_j|\theta_{j,g}) = \theta_{j,g} \mathbb{1}_{\{g=x_j\}}$$

Including all categories of the feature means adding G base learners $b_{j,1}, \dots, b_{j,G}$



HANDLING OF CATEGORICAL FEATURES



Advantages of simultaneously handling all categories in CWB:

- Much faster estimation compared to using individual binary BLs
- Explicit solution of $\hat{\theta} = \arg \min_{\theta \in \mathbb{R}^G} \sum_{i=1}^n (y^{(i)} - b_j(x_j^{(i)} | \theta))^2$:

$$\hat{\theta}_g = n_g^{-1} \sum_{i=1}^n y^{(i)} \mathbb{1}_{\{x_j^{(i)}=g\}}$$

- For features with many categories we usually add a ridge penalty

HANDLING OF CATEGORICAL FEATURES

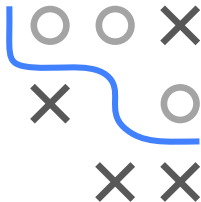
Advantages of including categories individually in CWB:

- Enables finer selection since non-informative categories are simply not included in the model.
- Explicit solution of $\hat{\theta}_{j,g} = \arg \min_{\theta \in \mathbb{R}} \sum_{i=1}^n (y^{(i)} - b_g(x_j^{(i)} | \theta))^2$ with:

$$\hat{\theta}_{j,g} = n_g^{-1} \sum_{i=1}^n y^{(i)} \mathbb{1}_{\{x_j^{(i)}=g\}}$$

Disadvantage of individually handling all categories in CWB:

- Fitting CWB is slower
- Penalization and selection become difficult since base learner has exactly one degree of freedom.



INTERCEPT HANDLING

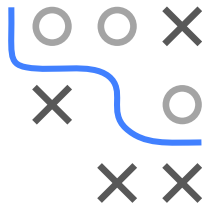
There are two options to handle the intercept in CWB. In both, the loss-optimal constant $f^{[0]}(\mathbf{x})$ is an initial model intercept.

❶ Include an intercept BL:

- Add BL $b_{\text{int}} = \theta$ as potential candidate considered in each iteration and remove intercept from all linear BLs, i.e., $b_j(\mathbf{x}) = \theta_j x_j$.
- Final intercept is given as $f^{[0]}(\mathbf{x}) + \hat{\theta}$. Linear BLs without intercept only make sense if covariates are centered (see [▶ Hofner et al. 2014](#) tutorial, p. 7)

❷ Include intercept in each linear BL and aggregate into global intercept post-hoc:

- Assume linear base learners $b_j(\mathbf{x}) = \theta_{j1} + \theta_{j2} x_j$. If base learner $\hat{b}_j$ with parameter $\hat{\theta}^{[1]} = (\hat{\theta}_{j1}^{[1]}, \hat{\theta}_{j2}^{[1]})$ is selected in first iteration, model intercept is updated to $f^{[0]}(\mathbf{x}) + \hat{\theta}_{j1}^{[1]}$.
- During training, intercept is adjusted M times to yield $f^{[0]}(\mathbf{x}) + \sum_{m=1}^M \hat{\theta}_{j1}^{[m]}$

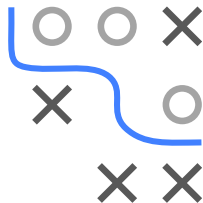


EXAMPLE: LIFE EXPECTANCY

Consider the `life_expectancy` data set (WHO, available on

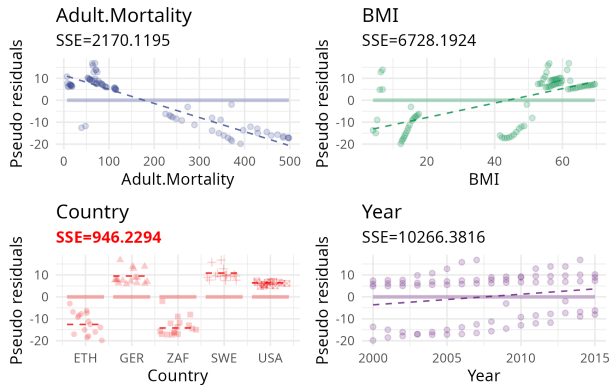
[Click for source](#)) : regression task to predict life expectancy.

We fit a CWB model with linear BLs (with intercept)



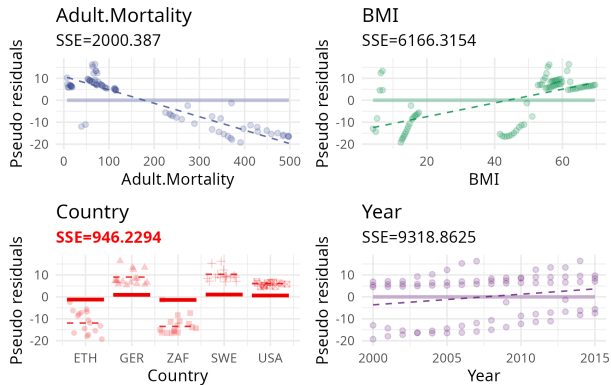
variable	description
<code>Life.expectancy</code>	Life expectancy in years
<code>Country</code>	The country (just a selection GER, USE, SWE, ZAF, and ETH)
<code>Year</code>	The recorded year
<code>BMI</code>	Average BMI = $\frac{\text{body weight in kg}}{(\text{Height in m})^2}$ in a year and country
<code>Adult.Mortality</code>	Adult mortality rates per 1000 population

Using `compboost` with $M = 150$ iterations, we can visualize which BL was selected when and how the estimated feature effects evolve over time.



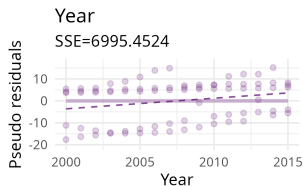
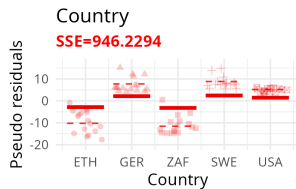
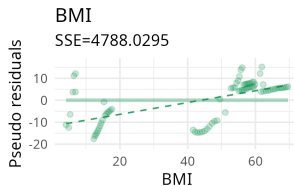
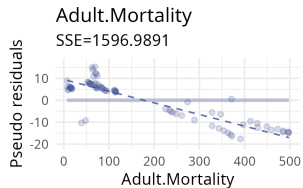
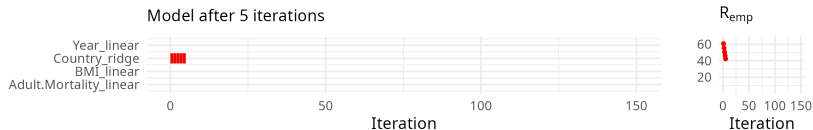
©

A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path consists of the following cells: (0,0), (0,1), (1,1), (1,2), and (2,2). The cells (0,2), (1,0), and (2,0) are empty. The cells (0,1), (1,1), and (1,2) contain a grey 'X'. The cells (1,0) and (2,0) contain a grey circle.

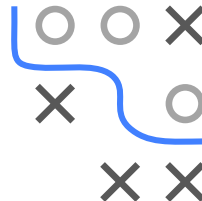


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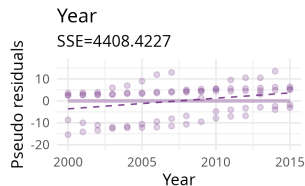
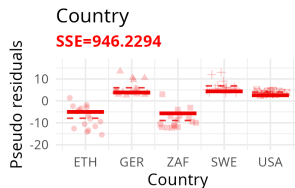
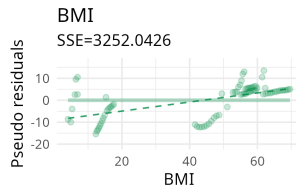
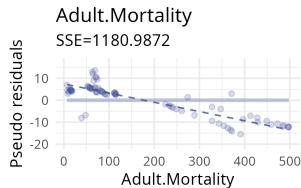
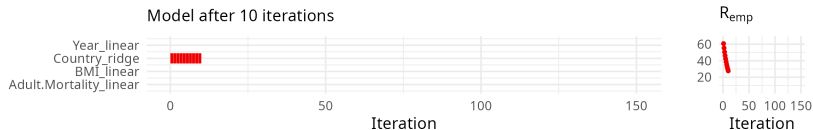
EXAMPLE: LIFE EXPECTANCY



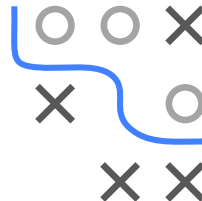
— Partial feature effect - - - - Base learner fit to pseudo residuals

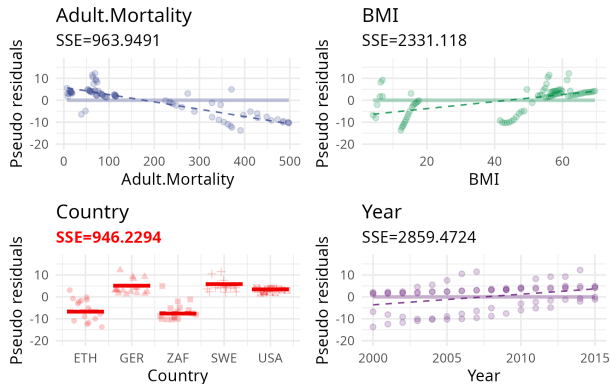


EXAMPLE: LIFE EXPECTANCY



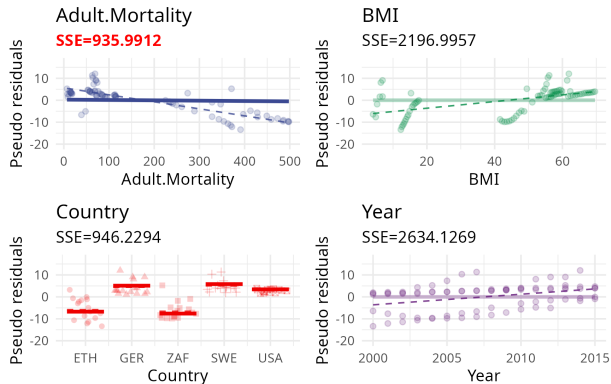
— Partial feature effect - - - Base learner fit to pseudo residuals





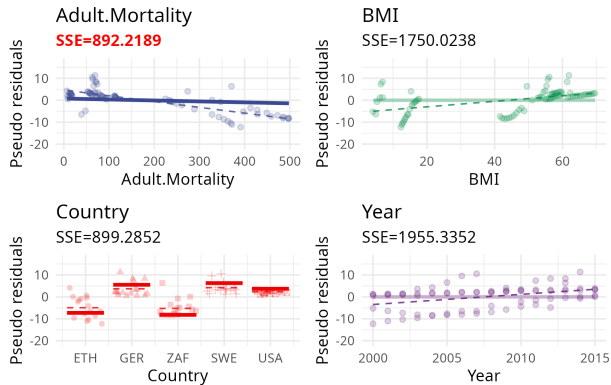
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A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path is composed of three segments: a horizontal segment from (0,0) to (0,1), a vertical segment from (0,1) to (1,1), and a horizontal segment from (1,1) to (2,1). The cells (0,2), (1,0), (1,2), and (2,0) are empty. The cells (0,1), (1,1), and (2,1) are occupied by a blue path.



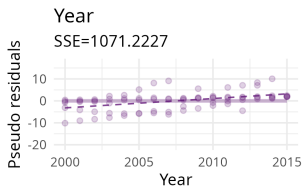
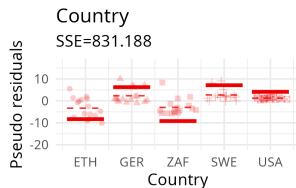
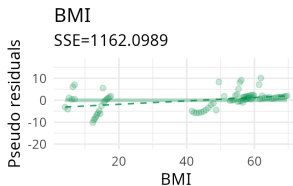
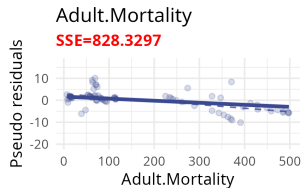
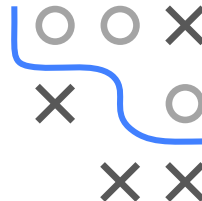
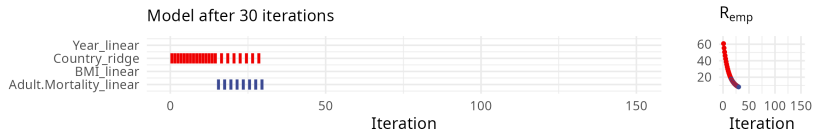
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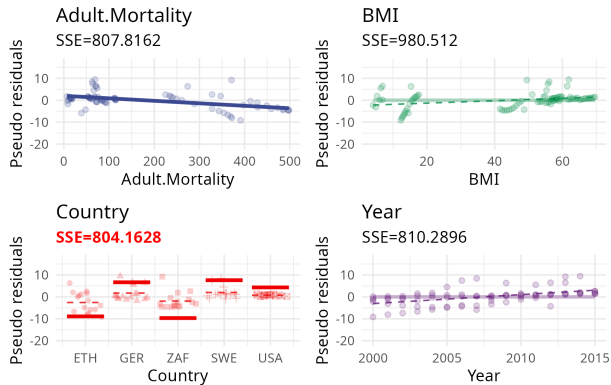


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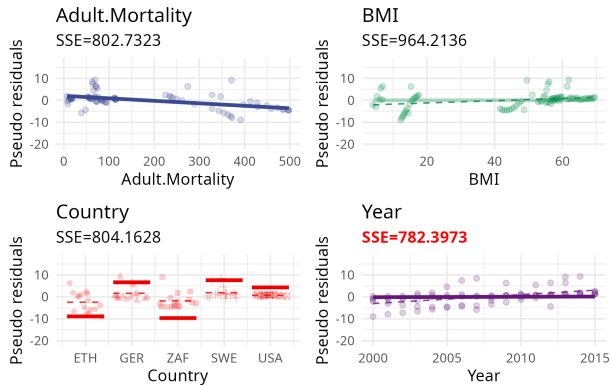
EXAMPLE: LIFE EXPECTANCY



— Partial feature effect - - - - Base learner fit to pseudo residuals

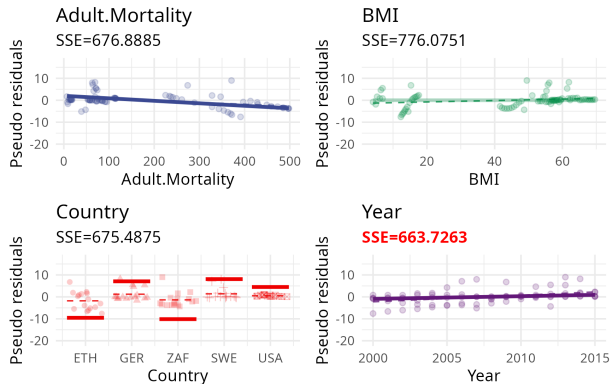
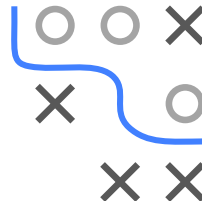
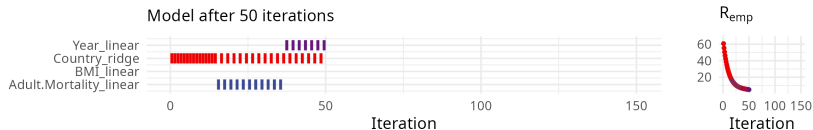


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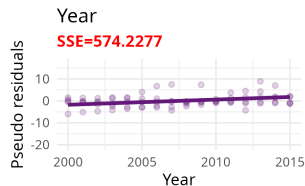
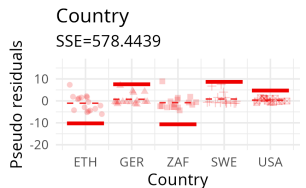
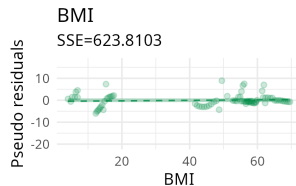
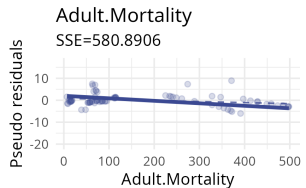
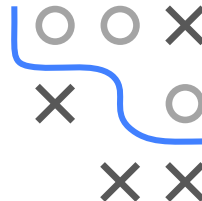
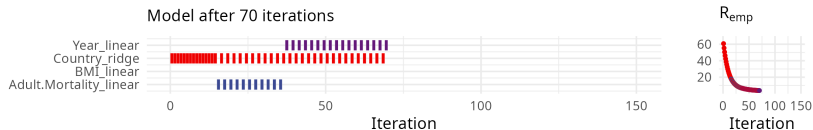
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EXAMPLE: LIFE EXPECTANCY



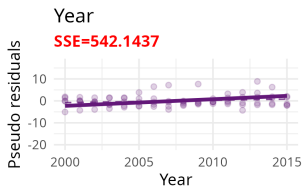
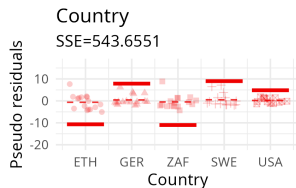
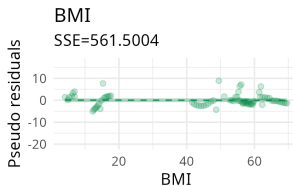
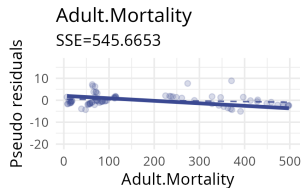
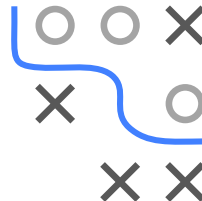
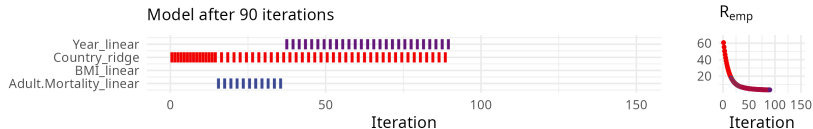
— Partial feature effect - - - - Base learner fit to pseudo residuals

EXAMPLE: LIFE EXPECTANCY

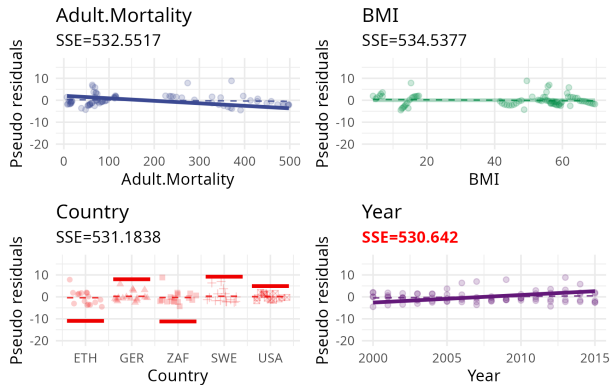


— Partial feature effect - - - - Base learner fit to pseudo residuals

EXAMPLE: LIFE EXPECTANCY

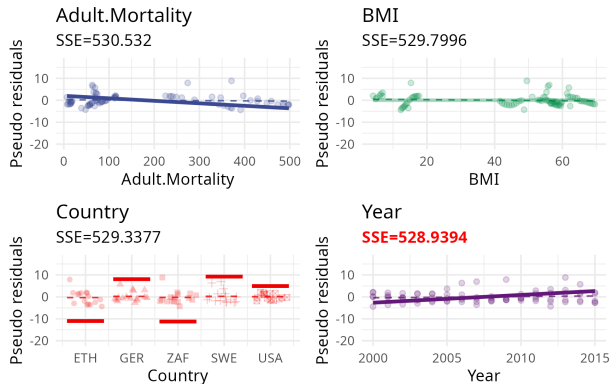
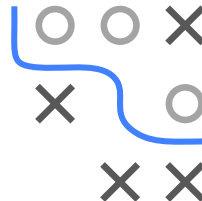
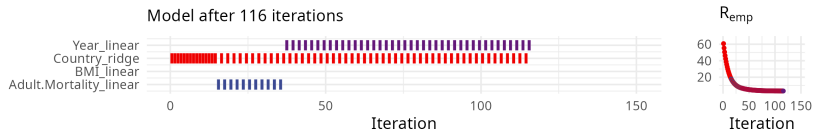


— Partial feature effect ···· Base learner fit to pseudo residuals



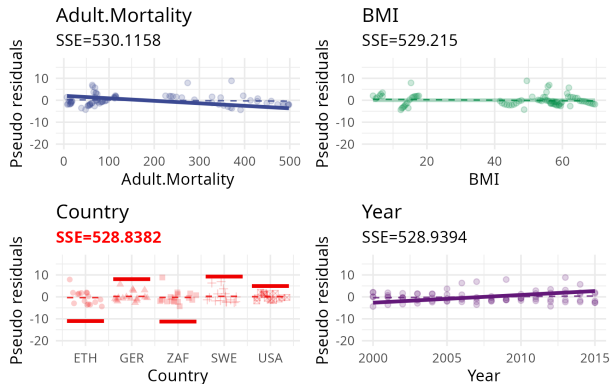
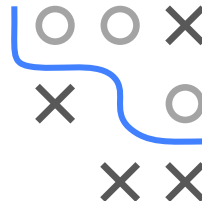
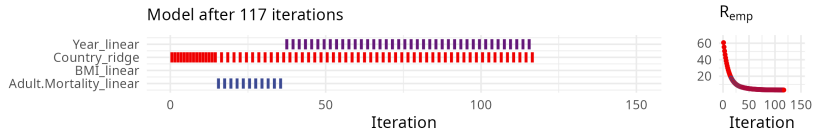
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EXAMPLE: LIFE EXPECTANCY



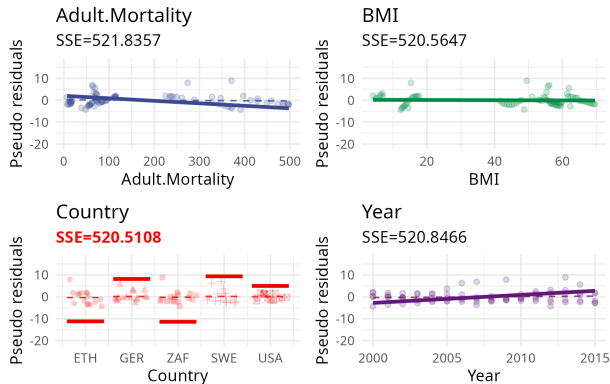
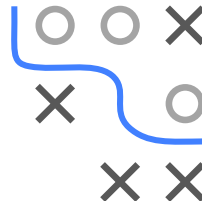
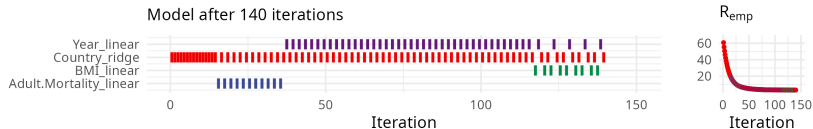
— Partial feature effect ···· Base learner fit to pseudo residuals

EXAMPLE: LIFE EXPECTANCY



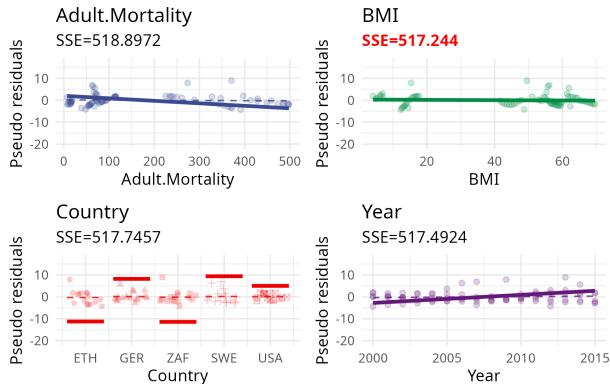
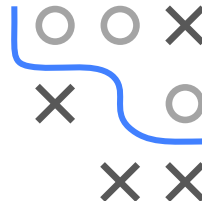
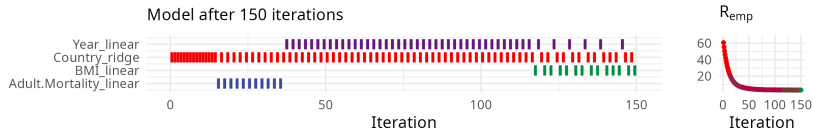
— Partial feature effect - - - - Base learner fit to pseudo residuals

EXAMPLE: LIFE EXPECTANCY



— Partial feature effect - - - - Base learner fit to pseudo residuals

EXAMPLE: LIFE EXPECTANCY



— Partial feature effect - - - - Base learner fit to pseudo residuals